

Vikram Kamath Cannanure

POSTDOCTORAL FELLOW - SAARLAND UNIVERSITY · HCI, ICTD, AI, EDUCATION

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I design *Thoughtful AI*: human-centered systems for reflection, care, and trust, built with communities across HCI and ICTD.

- **Research** — 18 full papers at top-tier HCI venues (CHI, CSCW, COMPASS, ICTD); systems, datasets, evaluation frameworks.
- **Impact** — AI systems deployed across the Global South; teacher chatbot in Côte d'Ivoire: 400+ teachers, 10,000+ students.
- **Funding** — \$215,000+ in competitive grants, including a Gates Foundation Grand Challenges award.
- **Recognition** — 509 citations, h-index 12 (Google Scholar, 2026); Best Paper Award at ACM COMPASS.
- **Leadership** — Co-led *HCI Across Borders*, a global HCI community; mentored 25+ students; reviewed 80+ papers.

Research Experience

Saarland University

POSTDOCTORAL FELLOW

Saarbrücken, Germany

Oct 2023 – present

- Host: Prof. Ingmar Weber · Leading a research area at the intersection of HCI, ICTD, and generative AI
- Co-supervising PhD and master's students on socially grounded AI projects
- 🤝 Co-PI on a \$100,000 Gates Foundation grant to co-design a chatbot improving women's and youth health in rural India
- Instructor for the graduate seminar "AI and the Global South"
- Key publications: [P13, P12, P14, S4]

Education

Carnegie Mellon University

PhD, HUMAN-COMPUTER INTERACTION

Pittsburgh, USA

2017–2023

- Advisors: Prof. Tim Brown, Prof. Amy Ogan · Committee: Chinmay Kulkarni, Mayank Goel, Kentaro Toyama (U. Michigan)
- Dissertation: *DIA: Supporting Teacher Professional Development in Low Infrastructure Settings*
- Led multi-year fieldwork and chatbot deployments for teachers in rural Côte d'Ivoire
- 🏆 Awarded \$100,000 Fellowship from the Center for Machine Learning and Health (CMLH)
- Key publications: [P6, P7, P8, P9, P10, P15]

Carnegie Mellon University

MS, HUMAN-COMPUTER INTERACTION

Pittsburgh, USA

2017–2021

Carnegie Mellon University

MS, EDUCATIONAL TECHNOLOGY AND APPLIED LEARNING SCIENCE

Pittsburgh, USA

2014–2015

National Institute of Technology Karnataka

B.TECH.

Karnataka, India

2007–2011

Grants and Awards (\$215,000+ total)

2026–	Listen at Scale (Gram Vaani) , Voice-first AI for reproductive and mental wellness — \$5,000	India
2025–	Gates Foundation Grand Challenges (Co-PI) , LLM wellness platform for rural India — \$100,000	India
2019	PhD Fellowship , Center for Machine Learning and Health, CMU — \$100,000	Pittsburgh, USA
2023, 2025	SIGCHI Development Grant , HCI Across Borders (HCIxB) workshop at CHI — \$10,000	
2019	Best Paper Award , ACM COMPASS	Accra, Ghana
2019	Poster Spotlight , ACM COMPASS	Accra, Ghana

Publications

- **Record** — 35+ peer-reviewed publications: 18 full papers at *CHI*, *COMPASS*, *ICTD*, *CUI*; 3 under review.
- **Metrics (2026)** — 509 citations, h-index 12, i10-index 15. [\[Google Scholar\]](#) [\[ORCID\]](#)
- **Author order** — first = primary intellectual lead; last = supervision and project leadership; * = equal contribution.

JOURNAL AND CONFERENCE FULL PAPERS [P]

- [P18] **Cannanure, V.K.**, Yinkfu, B., Bryan, D., Amin, M., Weber, I., *Teacher Professional Development on WhatsApp with LLMs: Early Lessons from Cameroon*. Accepted at Proceedings of the International Conference on Artificial Intelligence in Education (AIED 2026), Springer LNCS. [\[Link\]](#)
- [P17] Behmanush, H., Akhtari, F., Weber, I., **Cannanure, V.K.**, *Designing Safe and Accountable GenAI as a Learning Companion with Women Banned from Formal Education*. Accepted at Proceedings of the ACM Conference on Fairness, Accountability, and Transparency (FAccT 2026), ACM. [\[Link\]](#)

- [P16] Wolf, S.*, Ogan, A.*, Jasińska, K.*, Kembou, S.*, **Cannanure, V.K.**, Whitehead, H., Corbett, J., Tanoh, F., Wortsman, B., Lim, J., Zhao, H., Kumar, A. *Impacts of Targeted Instruction with Mobile-based Support on Teachers and Students in Rural Côte d'Ivoire*. Accepted at American Educational Research Journal. [\[Link\]](#)
- [P15] **Cannanure V.K.**, Ngoon T., Wolf S., Jasińska K., Brown T.X., Ogan A. *A Design-Based Research Approach to Bridge Teacher Aspirations and Goal-Setting*. *ACM Journal on Computing and Sustainable Societies*. [\[Link\]](#)
- [P14] Zhu J., Kempermann M., **Cannanure V.K.**, Hartland A., Navarrete R., Carteny G., Braun D., Weber I. *Learn, Explore and Reflect by Chatting: Understanding the Value of an LLM-Based Voting Advice Application*. *ACM CUI 2025*, Waterloo, Canada. [\[Link\]](#)
- [P13] Adhikari D.M., Hartland A., Weber I., **Cannanure V.K.**. *Exploring LLMs for Automated Generation and Adaptation of Questionnaires*. *ACM CUI 2025*, Waterloo, Canada. [\[Link\]](#)
- [P12] Behmanush H., Akhtari F., Nooripoor R., Weber I., **Cannanure V.K.** *Online Learning and GenAI: Supporting Women's Aspirations Amid Socio-Political Instability in Afghanistan*. *ACM COMPASS 2025* [\[Link\]](#)
- [P11] Verma R., Weber I., **Cannanure V.K.**. *Exploring AI in Education: Child-Computer Interaction in the Global South*. *ACM Journal on Computing and Sustainable Societies*. [\[Link\]](#)
- [P10] **Cannanure V.K.**, Ngoon T., Wolf S., Jasińska K., Brown T., Ogan A. *Understanding the Longitudinal Impacts of a Chatbot to Facilitate a Virtual Community of Practice for Teachers in Rural Côte d'Ivoire*. *ACM COMPASS 2024*, Delhi, India. [\[Link\]](#)
- [P9] Ngoon T., **Cannanure V.K.**, Wolf S., Jasińska K., Brown T., Ogan A. *"I believe I did not preach into the desert": Opportunities & Challenges in Scaling Teacher Mentorship through Mobile Technology in Rural Côte d'Ivoire*. *L@Scale 2024*, Atlanta, USA. [\[Link\]](#)
- [P8] **Cannanure V.K.**, Ávila E., Ngoon T., Adjii Y., Wolf S., Jasińska K., Brown T., Ogan A. *"We dream of climbing the ladder, for getting there, we have to do our job better": Designing for Teacher Aspirations in Rural Côte d'Ivoire*. *ACM COMPASS 2022*, Seattle, USA. [\[Link\]](#)
- [P7] **Cannanure V.K.**, Brown T., Ogan A. *DIA: A Human-AI Hybrid Chatbot for Low-Resource Contexts*. *ICTD 2020*, Guayaquil, Ecuador. [\[Link\]](#)
- [P6] **Cannanure V.K.**, Souvenir J., Adjii Y., Apke H., Jasińska K., Brown T., Ogan A. *"I'm fine where I am, but I want to do more": Exploring Teacher Aspirations in Rural Côte d'Ivoire*. *ACM COMPASS 2020*, Guayaquil, Ecuador. [\[Link\]](#)
- [P5] Madaio M., Yarzebinski E., **Cannanure V.K.**, Zinszer B., Seri A., Tanoh F., Hannon-Cropp J., Cassell J., Jasińska K., Ogan A. *Collective Support and Independent Learning with a Voice-Based Literacy Technology in Rural Communities*. *CHI 2020*, Hawaii, USA. [\[Link\]](#)
- [P4] 🏆 Madaio M., **Cannanure V.K.**, Yarzebinski E., Zasacky S., Tanoh F., Hannon-Cropp J., Cassell J., Jasińska K., Ogan A. *"You Give a Little of Yourself": Family Support for Children's Use of an IVR Literacy System*. *ACM COMPASS 2019*, Accra, Ghana. (Best Paper Award) [\[Link\]](#)
- [P3] **Cannanure V.K.**, Brown T. *RASP-IVR: A Low-Cost Interactive Voice Response System*. *AfriCHI 2018*, Windhoek, Namibia. [\[Link\]](#)
- [P2] Onafuwa D., Early K., Vyas N., **Cannanure V.K.**, Mankoff J. *Understanding the Information Needs of Prospective Tenants*. *ACM COMPASS 2018*, San Francisco, USA. [\[Link\]](#)
- [P1] Bhat K.*, Thinayane M.*, Goldkind L., **Cannanure V.K.** *Critical Participatory Design: Reflections on Engagement and Empowerment in a Case of a Community-Based Organization*. *Participatory Design Conference (PDC 2018)*, Hasselt and Genk, Belgium. [\[Link\]](#)

BOOK CHAPTERS [B]

- [B1] Gautam A., Wong-Villacrés M., Gaytán Lugo L.S., **Cannanure V.K.**, Vannini S., Jimenez A., Tuli A., Reynolds-Cuellar P., Pei L., Alvarado Garcia A., Badillo-Urquiola K., Prietch S.S., Densmore M., Banganayi T., Dray S. *Human-Computer Interaction Across Borders: Perspectives from the Global South*. In *The Human-Computer Interaction Handbook* (4th ed.), Julie A. Jacko (Ed.). CRC Press, 2026.

SHORT PAPERS AND POSTERS [S]

- [S5] **Cannanure V.K.**, Wolf S., Jasińska K., Brown T., Ogan A. *Enabling a Virtual Community of Practice Facilitated by a Chatbot for Rural Teachers*. *Proceedings of the 5th African Human-Computer Interaction Conference (AfriCHI 2025)*, Cairo, Egypt. [\[Link\]](#)

- [S4] Mensah E.E., **Cannanure V.K.**, Weber I. *Ethnic Diversity and Spatial Dynamics in Google Maps Reviews: Insights from a German Border City*. ICWSM 2025 (poster). [\[Link\]](#)
- [S3] **Cannanure V.K.**, Wolf S., Jasińska K., Brown T.X., Ogan A. *Applying Think-Aloud in ICTD: A Case Study of a Chatbot Use by Teachers in Rural Côte d'Ivoire*. ICTD 2024, Nairobi, Kenya. [\[arXiv\]](#)
- [S2] Adhikari D.M., **Cannanure V.K.**, Hartland A., Weber I. *Exploring LLMs for Automated Pre-Testing of Cross-Cultural Surveys*. ICTD 2024, Nairobi, Kenya. [\[arXiv\]](#)
- [S1] **Cannanure V.K.**, Chen X., Mankoff J. *Twist and Knock: A One-Handed Gesture for Smart Watches*. *Proceedings of the 42nd Graphics Interface Conference (GI 2016)*, Victoria, Canada. [\[Link\]](#)

WORKSHOP PAPERS AND LIGHTLY REVIEWED WORK [L]

- [L4] **Cannanure V.K.**, Varghese D., Rivera-Loaiza C., Noor F., Das D., Jain P., Chang-Rizzo M., Wong-Villacrés M., Karusala N., Ahmed N., Till S., Akhigbe B., Dray S., Densmore M., Sturm C., Kumar N. (2023). *HCI Across Borders: Towards Global Solidarity*. CHI '23, Apr 2023, Hamburg, Germany.
- [L3] **Cannanure V.K.**, Karusala N., Rivera-Loaiza C., Prabhakar A. S., Varanasi R. A., Tuli A., Gamage D., Noor F., Nemer, D., Das D., Dray S., Sturm C., Kumar, N. (2022). *HCI Across Borders: Navigating Shifting Borders at CHI*. CHI '22, Apr 2022, New Orleans, USA.
- [L2] **Cannanure V.K.**, Gamage D., Sturm C., Winschiers-Theophilus H., Maestre, J. F., Karusala N., Reynolds-Cuéllar P., Kumar, N., *Workshop: Decolonizing HCI Across Borders*. CHI '21, Apr 2021, Yokohama, Japan.
- [L1] Kumar N., **Cannanure V.K.**, Gamage D., Prabhakar A., Sturm C. *Workshop: HCI Across Borders and Sustainable Development Goals*. CHI '20, Apr 2020, Hawaii, USA.

Teaching Experience

AI and the Global South (Block Seminar)

LEAD INSTRUCTOR AND COURSE CREATOR

Student evaluation: 1.22 overall vs. department average 1.51 (lower is better)

Saarland University

Summer 2025

Elements of Data Science and Artificial Intelligence

COURSE COORDINATOR (INSTRUCTOR: PROF. INGMAR WEBER)

Saarland University

Fall 2024

Programming Usable Interfaces (05-430/05-630)

TEACHING ASSISTANT, HYBRID (INSTRUCTOR: PROF. SCOTT HUDSON)

Carnegie Mellon University

Spring 2022

User-Centered Research & Evaluation (05-410/05-610)

TEACHING ASSISTANT, REMOTE (INSTRUCTOR: RAELEN MUSURACA)

Carnegie Mellon University

Spring 2021

Mentoring

PhD Students (2)

***Hamayoon Behmanush**: Saarland University, PhD: *Supporting Afghan women in learning programming* ([P12](#), [P17](#))

***Ethel Mensah**: Saarland University, PhD: *Exploring structural inequality in Germany using POI data* ([S4](#))

Thesis Committees (1)

Ahana Ghosh: MPI-SWS, PhD 2026: committee member

Master's Students (21)

***Divya Mani Adhikari**: Saarland University, MS 2025: *Automated Survey Testing using LLMs* ([P13](#), [S2](#))

Prithika Acharya: CMU Pittsburgh, METALS 2022: *Teacher Training Project in Côte d'Ivoire*

Aastha Patel: CMU Pittsburgh, METALS 2022: *Teacher Training Project in Côte d'Ivoire*

Egide Kamuzinzi: CMU Africa, MSIT 2021: *Teacher Training Project in Côte d'Ivoire*

Baptiste Moreau-Pernet: CMU Pittsburgh, METALS 2021: *Teacher Training Project in Côte d'Ivoire*

Saumya Sharma: CMU Pittsburgh, MSIIPM 2020: *Simple Health Information System*

Lucie Uwimpaye: CMU Africa, MSIT 2021: *Simple Health Information System*

Eloísa Ávila: CMU Pittsburgh, MSPPM 2021: *Teacher Training Project in Côte d'Ivoire*

Sheila Mbadi: CMU Africa, MSIT 2021: *Simple Health Information System*

Patrick Manirakiza: CMU Africa, MSECE 2019: *M&E system for Rwanda Women's Network*

Solange Iyubu: CMU Africa, MSECE 2019: *M&E system for Rwanda Women's Network*

Sabin Sheja: CMU Africa, MSECE 2019: *M&E system for Rwanda Women's Network*

Justin Bigwi: CMU Africa, MSECE 2019: *Design of a Phone Call-Based Health Information System*
Gloria Umutesi: CMU Africa, MSECE 2020: *Design of a Phone Call-Based Health Information System*
Jocelyne Mukamisha: CMU Africa, MSIT 2020: *Design of a Phone Call-Based Health Information System*
Justin Souvenir Niweteto: CMU Africa, MSIT 2020: *Design of a Phone Call-Based Health Information System* (P6)
Fraterne Kagame: CMU Africa, MSIT 2020: *Design of a Phone Call-Based Health Information System*
Huguette Ingenere: CMU Africa, MSIT 2020: *French interview transcription for teachers in Côte d'Ivoire*
Marie Claudine Ingenere: CMU Africa, MSIT 2020: *French interview transcription for teachers in Côte d'Ivoire*
Nebiyu Yismaw: CMU Africa, MSECE 2018: *Digital and voice-based systems to support gender-based violence response*
Fabiola Uwashema: CMU Africa, MSIT 2018: *Digital and voice-based systems to support gender-based violence response*

Undergraduate Students (3)

***Riti Verma:** Indian Institute of Technology, Delhi: *Exploring the role of AI in the Global South* (P11)
Lena Armstrong: University of Pennsylvania, Summer REU: *Teacher Training Project in Côte d'Ivoire*
Maya Greenholt: CMU Pittsburgh, Cognitive Science: *Teacher Training Project in Côte d'Ivoire*

*Mentored during my postdoctoral fellowship at Saarland University.

Professional Service

CHAIRING AND ORGANIZING

2026	Papers Chair , ACM COMPASS 2026	Virtual
2025	Lead Mentor , HCI Across Borders (HCIXB) workshop at CHI 2025	Yokohama, Japan
2024	Lead Organizer , HCI Across Borders (HCIXB) workshop at ACM COMPASS	Delhi, India
2023	Lead Organizer , HCI Across Borders (HCIXB) workshop at CHI	Hamburg, Germany
2022	Lead Organizer , HCI Across Borders (HCIXB) workshop at CHI	New Orleans, USA
2021	Lead Organizer , HCI Across Borders (HCIXB) workshop at CHI	Yokohama, Japan
2020	Organizer , HCI Across Borders (HCIXB) symposium at CHI	Hawaii, USA
2019	Organizer , HCI Across Borders (HCIXB) symposium at CHI	Glasgow, UK
2019	Organizer , Young Researchers Reception at ICTD	Ahmedabad, India
2020–2022	Executive Committee Member , SIGCHI Development Fund	

PROGRAM COMMITTEES AND REVIEWING

2024– **Program Committee**, ACM COMPASS 2025, AfriCHI 2025, ICTD 2024
 2020–2023 **Associate Chair**, CHI Extended Abstracts
 2018– **Reviewer (80+ papers)**, CHI, CSCW, ACM COMPASS, ICTD, CHI Case Studies

Professional Experience

Carnegie Mellon University, Africa

VISITING RESEARCHER

- Host: Prof. Tim Brown
- Project: Design of a Phone Call-Based Health Information System
- Conducted participatory research with rural women and prototyped an IVR system in rural Rwanda

United Nations University Computing and Society

RESEARCH INTERN

- Advisor: Dr. Mamello Thinyane
- Conducted participatory design with a local NGO
- Built a mobile app to report and monitor homelessness in Macau

Carnegie Mellon University

SENIOR WEB DEVELOPER

- Advisor: Prof. Geoff Gordon
- Built and managed educational technology to support online learning
- Prototyped a low-cost video recording system for teaching courses

Carnegie Mellon University

RESEARCH ASSISTANT

- Advisor: Prof. Jen Mankoff
- Longitudinal eating detection: deployed and evaluated the system with 12 users in the field
- Twist & Knock: a one-handed gesture for smartwatches (S1)
- EDigs: built and maintained the APIs for an apartment-hunting website in Pittsburgh

Kigali, Rwanda
 2018–2020 (summers, spring)

Macau, SAR, China
 Jul 2017 – Sep 2017

Pittsburgh, USA
 Jul 2015 – Jun 2017

Pittsburgh, USA
 Aug 2015 – Jun 2017

- Engineered APIs for Android, iOS, web, and Google Glass
- Scaled back-end to handle 4 million users and manage 100K locations
- Designed a 3D viewer on par with Google Street View
- Managed 10 Ubuntu machines with MySQL and Redis replication

Skills

Methods	Participatory design, longitudinal field deployment, interviews, think-aloud, mixed-methods analysis
Programming	Python, JavaScript, PHP, HTML/CSS, Android, iOS, Arduino
Tools and Platforms	LLM APIs, chatbot and IVR platforms, WhatsApp deployment, pandas, scikit-learn, MySQL, Redis

Selected Press Coverage

The Next Web — *TeliportMe introduces 3D effects to its 360-degree panoramic app for Android*

TechCrunch — *Test Flighting: Applover Launches Free, Crowdsourced Testing Platform for Android Apps*

Robert Scoble — *Developers: crowd-source your testing of Android phones*

CMU — *A new online resource from Carnegie Mellon University seeks to help people find the perfect rental*

CMU HCII — *EDigs Helps Apartment Hunters Find the Perfect Rental*

The Better India — *How a National Level Cyclist Is Fighting Both Spinal Injury & Poverty with Inspiring Confidence*

Milaap Blog — *Getting Back to Cycling: A National Medalist's Dream*

Beyond Research

- **Philanthropy** — Raised over **\$20,000** for causes in India through swimming and crowdfunding
- **Android** — Shipped 3 apps to the Play Store, with roughly 10,000 active users
- **Web** — Deployed 20+ websites to production; the largest reached over **4 million users**
- **Teaching** — Taught Kannada as a second language to children at the S.V. Temple in Pittsburgh
- **Swimming** — Swam 10 miles from Elephanta Caves to the Gateway of India to raise money for acid-attack survivors